MATH ASSIGNMENT.

AKELLO ESTHER OKALANY   25/U/BIO/023/GV

I wrote C program using DevC/C++ software for the assignment.

[*] mtc.c

```c
#include <stdio.h>
int main() {
    int n, a, sum = 0;  // declare variables
    // n = term number, a = value of term, sum = total sum
    // sum = 0 meaning starting from 0
    printf("n\t a_n\n");   // print table headers for n and a_n
    printf("----------------\n");

    //loop through n = 1 to 15
    for(n = 1; n <= 15; n++) {
        a = 4*n - 3;    // calculate the nth term using formula

        printf("%d\t %d\n", n, a);   // print term number and value in table format
        sum += a;       // add this term to the running sum
    }

    // after loop ends, print the total sum
    printf("----------------\n");
    printf("Sum of first 15 terms = %d\n", sum);

    return 0;
}
```

For part a) This is the output of the C program showing the first 15 terms of the sequence $a_n = 4n - 3$.

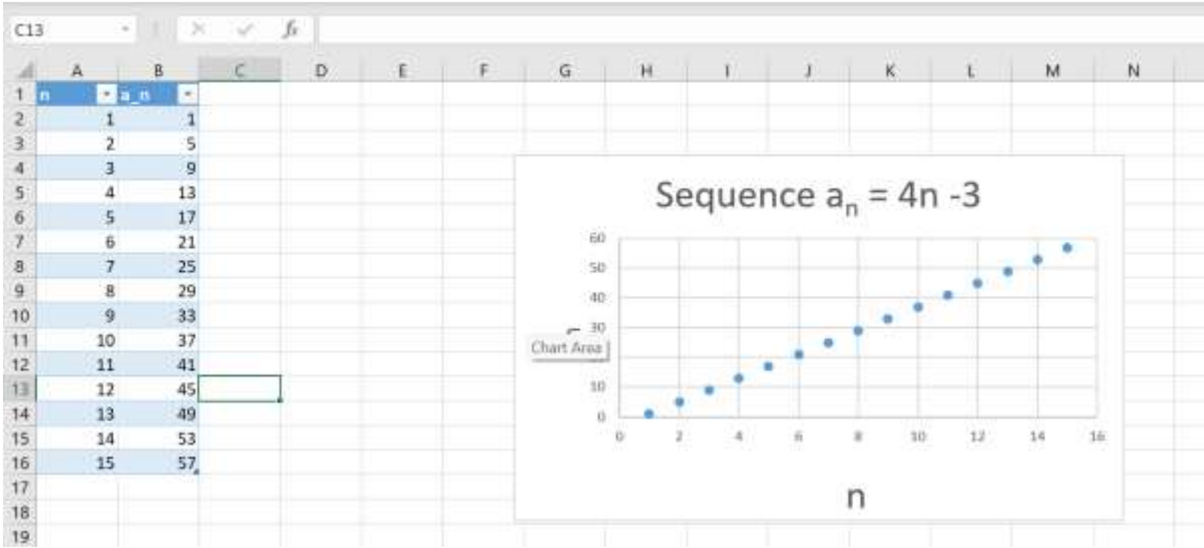
c) The program also computed the sum of the first 15 terms as shown;

C:\Users\User\Desktop\AKELLO-C\mtc.exe

```
n         a_n
----------------
1         1
2         5
3         9
4         13
5         17
6         21
7         25
8         29
9         33
10        37
11        41
12        45
13        49
14        53
15        57
----------------
Sum of first 15 terms = 435

--------------------------------
Process exited after 0.1946 seconds with return value 0
Press any key to continue . . .
```

b) The graph and the table, both using Excel.



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d)The explanation;

The sequence $a_n = 4n - 3$ is an arithmetic progression with first term $a_1 = 1$ and common difference, d = 4 from8 $a_{n+1} - a_n$ for any 2 consecutive numbers in the sequence.

Each term increases by 4 thus growth is steady, forming a straight-line pattern when plotted.

The sum of the first 15 terms is **465**, which can also be verified using the formula S_n = n/2 × (first term + last term) = 15/2 × (1 + 57) = 465.